SIMCODES:

AUTOMATING THE FRAGMENTATION OF PROTEINS

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PROJECT OVERVIEW

Accurately predicting enzymatic reactions requires reliable energy calculations for proteins, ligands, and their environment. Traditional quantum chemical methods are extremely **slow** and **computationally intensive** for large systems. This creates a need for faster approaches that balances speed with accuracy. Machine learning offers a promising solution to this challenge, particularly in automating the process of protein fragmentation.

Fragment Molecular Orbital (**FMO**) and Many-Body Expansion (**MBE**) are current methods used to get energy approximations, Our approach builds on the idea by generating protein fragments in a way that their total energy is very close to original protein's energy. Hence, while running energy calculations we can use expensive methods in parallel drastically reducing the overall time and complexity. Currently we're building the model for peptides level fragmentation and then plan to scale up to proteins, allowing a more iterable process for model fine-tuning.

As molecular structures resemble **graph like structure** and properties, we decided to use **graph clustering algorithms/models** to generate fragments. To make chemical sense we also cap the fragments with hydrogens and adjusted the final energy calculations according to caps added.

AFFILIATIONS

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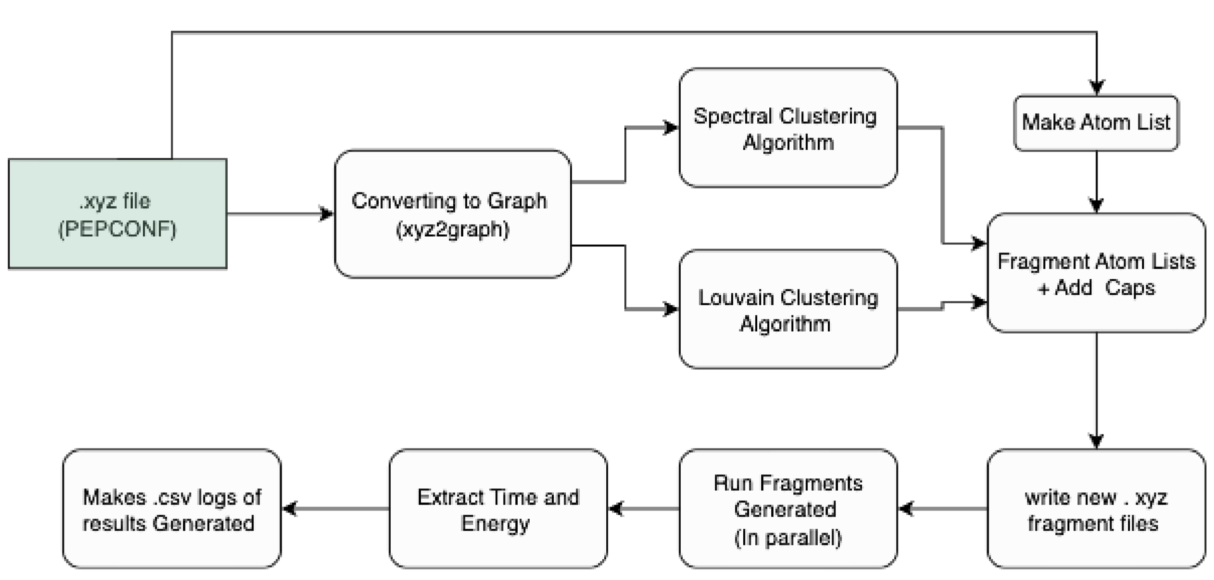
SOLUTIONS EXPLORED

We built a Python pipeline to accelerate accurate energy predictions for peptides by integrating packages including : xTB, NetworkX and xyz2graph:

1. **Graph-Based Fragmentation:** Molecules were represented as graphs, and using Spectral Clustering and Louvain Method to generate fragments. SC was more consistent and accurate.
2. **Hydrogen Capping Correction:** Hydrogens were added at cuts and corrected using half the energy of an H_2 molecule.
3. **Parallelization of xTB:** Fragment files were generated and processed with parallelized xTB jobs, reducing compute time.
4. **Hyperparameter Tuning:** tuned parameters like number of clusters, resolution, and threshold, evaluating them with scatter plots and heatmaps. Lower cluster counts typically produced better energy estimations. Also, there's a clear trade-off between fragment count and energy deviation.
5. **Scalability :** Aim is to achieve <0.01 Hartree error and scale to proteins once the method is validated on peptide dataset.

CURRENT PIPELINE



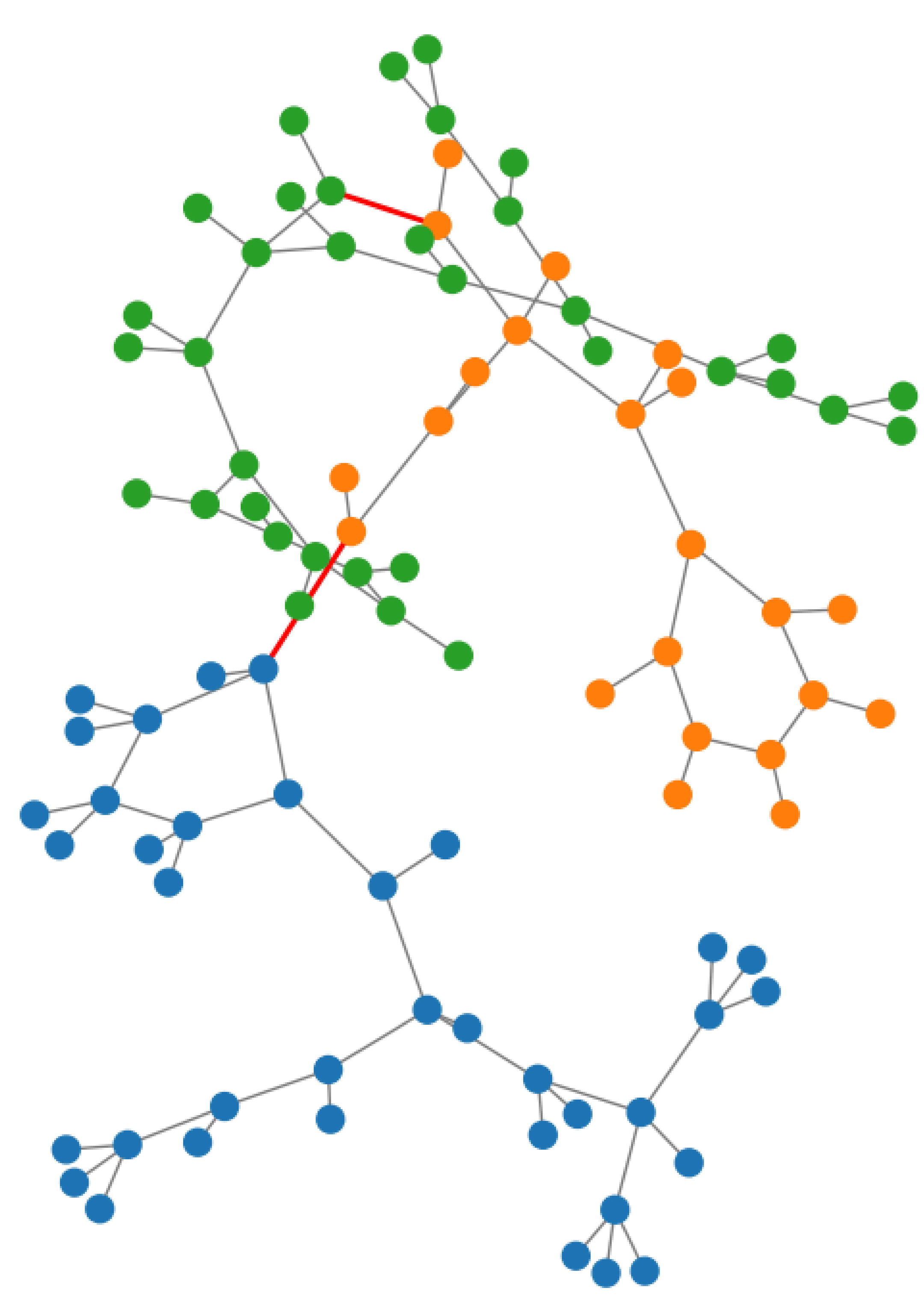
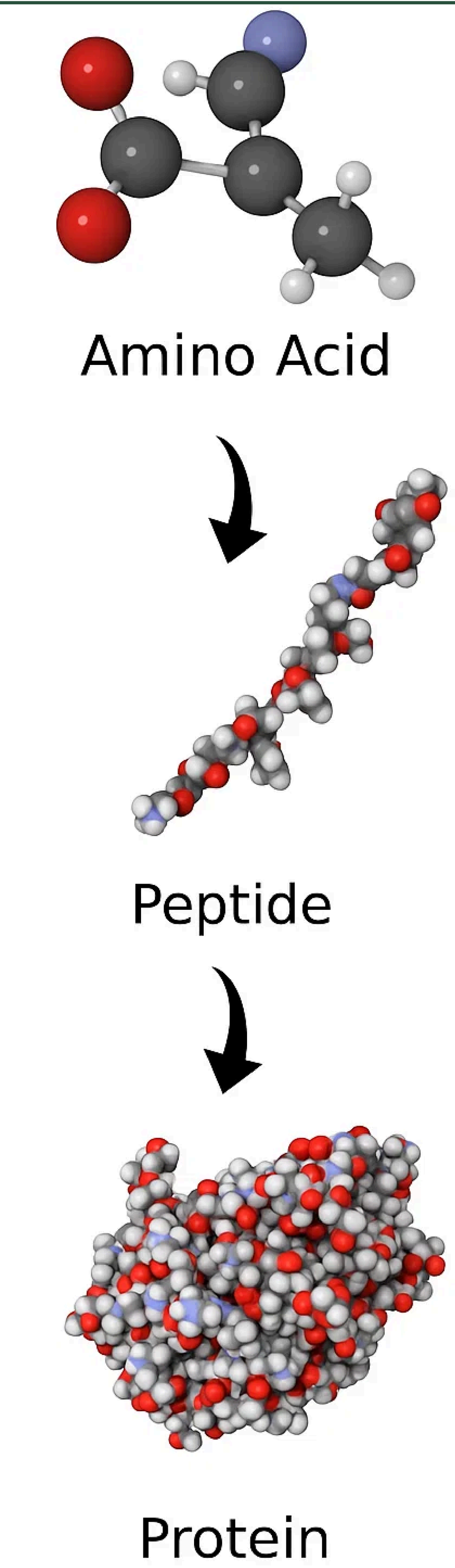


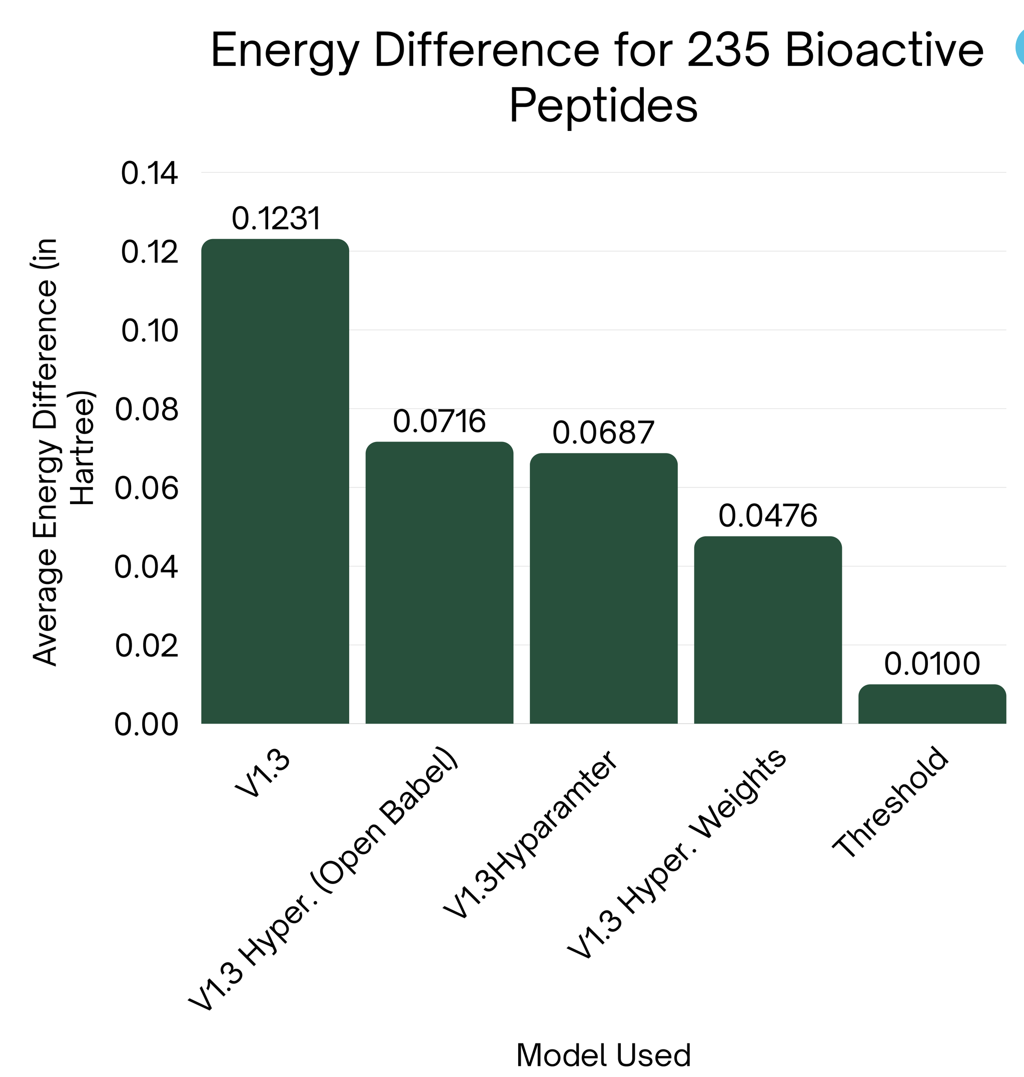
Photo of Fragmented networkX graph made using NetworkX

PROGRESS MADE

- Validated that hydrogen capping preserves energy consistently, by manually capping and fragmenting peptides and running xTB calculations.
- Implemented Spectral Clustering and Louvain Method to fragment peptides and built a pipeline that generates fragment files, runs xTB via subprocess, and logs results in .csv files.
- Parallelized xTB energy calculations, significantly reducing runtime for large peptide batches.
- Developed a hyperparameter tuning module to test different fragmentation settings and log optimal clustering configurations.
- Tuned and ran Spectral Clustering on over 2000 peptides, corrected redundancies and visualized how clustering parameters affect energy difference.
- Added chemically-aware weights to graph edges using bond types (i.e., penalizing double/triple bond breaks), and experimented using valency rules and Open Babel API to improve energy prediction accuracy.



Shows how training on peptides will help generalize to proteins.
source: https://www.peptidesciences.com/glossary/peptides-vs-proteins/



Average Time Reduced in Energy Calculation after fragmentation

CHALLENGES

1. **Limited data availability:** makes it challenging to train data-driven ML models and generating new data is time and resource intensive.
2. **Balancing chemical accuracy with abstraction:** Simplifying molecular systems for computational feasibility while maintaining chemically accurate fragmentation.
3. **Combinatorial complexity of fragmentation:** Exhaustively exploring all possible fragmentations is computationally infeasible, so finding optimal fragmentation is not possible.

ANALYSIS

The models ran over 2,500 peptides, showing that using pre-trained fragmentation models on protein graphs gave an error of **0.1231 Hartree**. Further hyperparameter tuning brought the error down to **0.0687**, reducing it by nearly half. Incorporating chemistry-aware edge weights reduced this error to **0.0476**, marking a 61% improvement from the original error. In contrast, using Open Babel valency weights increased the error.

Meanwhile, parallel execution of fragment-based xTB calculations resulted in a **95%** reduction in runtime. It's equivalent to performing an hour long energy calculation in about 3 minutes.

Our goal is to further reduce the error to below **0.01 Hartree** and scale this pipeline to full protein fragmentation, offering a fast and accurate alternative for protein energy prediction.

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Github: https://github.com/SIMCODES-ISU/Devarsh_Repo